

Highlight of Skills

- IT skills: Programming languages (R, Python, SQL), QGIS, Tableau, Microsoft Office software
- Languages: English, Cantonese, Mandarin
- Experienced in working as a team with high reliability and self-motivation
- Strong problem-solving and critical thinking skills, attentive to details

Education

- Sep 2025 – **MSc in Statistics**, *University of Toronto*, Toronto, ON, Canada
Apr 2026 (expected) Courses: Applied Multivariate Analysis, Methods of Applied Statistics, Special Topics in Statistical Data Science, Special Topics in Applied Statistics, Survival Analysis
- Sep 2021 – **H.B.Sc in Statistical Science (Specialist), Mathematics (Major), Health Studies (Focus)**, *University of Toronto*, Toronto, ON, Canada **cGPA: 3.94/4.00**
Apr 2025 Courses: Applied Statistical Methods, Applied Spatial Analysis, Time Series Analysis, Surveys, Sampling and Observational Data, Statistical Consultation
Awards: Samuel Beatty In-course Scholarship (2025), University of Toronto Excellence Award (2024), Eleventh Annual Canadian Statistics Student Conference Undergraduate Level First Prize (2023), University of Toronto Dean's List Scholar (2021–2025)

Data-related Experiences

- Sep 2025 – **R-Based Learning Designer**, *University of Toronto Human Biology Program*
Current
 - Developed 4 testable R coding assignments, improving students' learning engagement and pedagogical effectiveness.
 - Designed 2 interactive LearnR modules content with guided coding exercises and visualizations using RShiny, improving skills in reproducible coding, user-centered design, and visualization best practices.
 - Collaborated with peer module developers to address issues and revise updated modules, strengthening teamwork and technical communication skills.
- Sep 2025 – **Statistical Neuroimaging Researcher**, *University of Toronto, Department of Statistical Sciences*
Dec 2025
 - Investigated statistical inference for genetic heritability of intermodal coupling using simulated multi-modal brain MRI data, providing insights for advanced methodological approaches in neuroimaging data.
 - Conducted 50+ large-scale simulation experiments and applied adaptive clustering techniques in R with parallel computing, validating proposed model's ability in controlling Type I error rates and statistical power.
 - Summarized findings in a detailed report, providing evidence that informed refinement of current methods and shaped the development of more robust neuroimaging statistical approaches.
- May 2024 – **Regression Analysis Researcher**, *University of Toronto Department of Statistical Sciences*
– Current
 - Conducted research on robust regression methods for handling outliers and heavy-tailed data, resulting in a preprint paper and a departmental research day presentation.
 - Developed and evaluated 4 parameter estimation approaches under high-dimensional and outlier settings, generating a comprehensive performance analysis that provided insights for improving traditional model accuracy.
 - Implemented and troubleshoot optimization algorithms to address non-convergence issues and enhanced computational efficiency, enhancing proficiency for statistical computing through utilizing software R and Python.
 - Presented research findings through detailed reports and visualizations and published an R package with 20+ reusable functions, strengthening statistical communication and facilitate research reproducibility.
- Sep 2024 – **Machine Learning Researcher**, *University of Toronto Department of Statistical Sciences*
Apr 2025
 - Investigated fairness in machine learning with biased labels in healthcare data extracted from MIMIC-III clinical database consisting of 40,000+ ICU patient records, leading to a departmental research day presentation.
 - Evaluated embeddings-building algorithms performance in capturing semantic relationships among 6,000+ medical concepts, uncovering predictors that enhanced downstream modeling.
 - Trained supervised regression and random forest models for depression diagnosis and conducted fairness evaluations across demographic groups, revealing disparities in prediction outcomes linked to biased label distributions.
 - Maintained a reproducible workflow in R and curated a GitHub repository to share code and results, strengthening transparency and enabling replication by other researchers.

Sep 2024 – **Data Visualization Analyst, University of Toronto Human Biology Program**

- Analyzed student learning resources engagement data from Quercus and course grades using Excel, results informed data-driven evidence-based teaching strategies for course improvement.
- Created 100+ Tableau visualizations that transformed complex datasets into clear, compelling insights for faculty, showcasing advanced data visualization proficiency and strong statistical communication skills.
- Collaborated with peers to troubleshoot data challenges and refined visualization approaches, strengthening problem-solving and teamwork skills.

Jun 2024 – **Psychiatric Education Quantitative Researcher, Centre for Addiction and Mental Health**

- Conducted research on a study investigating relationship between physical movement and cybersickness symptoms in 2 immersive VR medical trainings, resulting in a peer-reviewed publication and 2 poster presentations.
- Summarized descriptive statistics and performed statistical analyses in R, uncovering a significant correlation between nausea severity and the extent of movement.
- Provided data-driven recommendations for the design of VR training programs that prioritize user comfort, emphasizing the importance of movement considerations while maintaining educational efficacy.

Sep 2023 – **Agent-Based Modeling Researcher, University of Toronto COBWEB lab**

- Initiated research using agent-based simulation to investigate declining birth rates and potential solutions, resulting in a poster presentation.
- Conducted comprehensive literature reviews and prepared an annotated bibliography of 30+ sources, synthesizing multidisciplinary findings that improved study design and strengthened critical thinking and organizational skills.
- Mastered the simulation software COBWEB to model population dynamics and policy scenarios, independently developing 20+ simulations that expanded research scope and demonstrated advanced self-learning ability.
- Performed time-series and regression analyses in Python to assess the impact of policy interventions on birth rates trends, providing insights for short-term and long-term strategic proposal.

May 2023 **Data Analyst Intern, Canadian Urban Institute**

- Developed regression models in R to analyze nationwide household and business-level carbon emissions and social connection patterns, providing data-driven insights for an ongoing Canadian city planning sustainability project.
- Created an interactive tool to demonstrate the effects of urbanization and housing composition on greenhouse gas emissions, enabling stakeholders to model scenarios for policy planning.
- Utilized geospatial visualization tool QGIS to produce 300+ detailed maps, combined with statistical findings, revealing urban planning insights and actionable recommendations for decision-makers.

Oct 2022 – **Computational Social Science Researcher, University of Toronto Population Well-being Lab**

- Applied machine learning methods (Gaussian Mixture Model, k-means, and hierarchical clustering) in R, to uncover meaningful subpopulations based on life satisfaction and demographics, resulting in 3 conference presentations.
- Prepared detailed 4 tip-sheets on algorithms and evaluated effectiveness and appropriateness of statistical methods, enhancing team understanding in methodological application.
- Designed a conference poster discussing the suitability of clustering methods and implications of results, effectively translating complex analyses into actionable insights for diverse audiences, resulting in a poster first prize.

Sep 2022 – **Statistical Consultant, University of Toronto Data Sciences Café**

- Contributed to a supportive and collaborative consulting community, fostering an environment where interdisciplinary research and statistical analysis could thrive.
- Recommended tailored statistical methodologies and identified limitations in proposed study designs to 20+ clients, delivering actionable insights that improved methodological validity and analytical outcomes across diverse fields.
- Identified potential biases and limitations in proposed research methods, providing actionable insights for improvement in study method design.
- Prepared 5 reports on advanced statistical topics, introducing statistical concepts and coding methods in an accessible manner to enhance statistical literacy among non-statistician researchers.

Other Experiences

Sep 2023 – **Teaching Assistant and Learning Support, University of Toronto Department of Statistical Sciences and**
Current **Data Sciences Institute**

DoSS Courses: STA130, STA238, STA304, Statistics Aid Centre, Florence Nightingale Day (2024, 2025)

DSI Modules: SQL, Applying Statistical Concepts, Sampling, Visualization

- Prepared tutorials and office hours for statistics courses, strengthening programming (R, Python, SQL) skills and addressed questions across a wide range of statistical courses, enhancing students' statistical learning.
- Engaged with high school students at events such as Florence Nightingale Day, translating high-level statistical concepts into accessible and engaging explanations to spark interest in statistics.

Sep 2022 – **First-year Learning Community Assistant Peer Mentor**, *University of Toronto*

- Apr 2023
- Facilitated 12 sessions to guide 20+ first year students in academic matters, fostering their skills to overcome challenges and manage transition anxiety to university life.
 - Established a rapport and inclusive community among students, promoting a supportive learning environment.
 - Coordinated discussions with Mathematics professors and library staff and managed online Quercus page to facilitate remote communication and engagement with students.

Volunteer

Sep 2022 – **Data Entry Volunteer**, *St. Michael's Hospital Trauma Research Program and Injury Prevention*

- Dec 2022
- Reviewed eligible medical charts to collect injury/poisoning information
 - Documented medical records into the system